HW5

2025-03-16

data: bread+basket

top 3 popular

```
head(sort(table(cafe$Item), decreasing = TRUE))
```

```
##
##   Coffee    Bread      Tea     Cake   Pastry Sandwich
##     5471     3325     1435     1025      856      771
```

```
coffee <- 1:length(caf_rule)
coffee <-
coffee[str_detect(caf_rule@rhs@itemInfo$labels[caf_rule@rhs@data@i+1]
,"Coffee")]

bread <- 1:length(caf_rule)
bread <- bread[str_detect(caf_rule@rhs@itemInfo$labels[caf_rule@rhs@data@i+1]
,"Bread")]

tea <- 1:length(caf_rule)
tea <- tea[str_detect(caf_rule@rhs@itemInfo$labels[caf_rule@rhs@data@i+1]
,"Tea")]
```

Coffee

```
inspect(head(sort(caf_rule[coffee], decreasing = TRUE),by = "lift",10))
```

```
##      lhs                           rhs               support confidence
coverage      lift count
## [1]  {period_day=morning,
##       weekday_weekend=weekday} => {Item=Coffee} 0.081874482  0.3245072
0.25230409 1.2163532  1679
## [2]  {period_day=morning}      => {Item=Coffee} 0.124884186  0.3047358
0.40981128 1.1422442  2561
## [3]  {weekday_weekend=weekday} => {Item=Coffee} 0.172770274  0.2766456
0.62451846 1.0369532  3543
## [4]  {period_day=morning,
##       weekday_weekend=weekend} => {Item=Coffee} 0.043009704  0.2730650
0.15750719 1.0235321   882
## [5]  {weekday_weekend=weekend} => {Item=Coffee} 0.094016677  0.2503896
0.37548154 0.9385377  1928
## [6]  {period_day=afternoon,
##       weekday_weekend=weekday} => {Item=Coffee} 0.087677378  0.2472157
0.35465938 0.9266410  1798
## [7]  {period_day=afternoon}    => {Item=Coffee} 0.137660311  0.2440142
0.56414883 0.9146406  2823
```

```
## [8]  {period_day=afternoon,
##       weekday_weekend=weekend} => {Item=Coffee} 0.049982933  0.2385940
0.20948944 0.8943243  1025
## [9]  {period_day=evening,
##       weekday_weekend=weekday} => {Item=Coffee} 0.003218413  0.1853933
0.01735993 0.6949113    66
## [10] {period_day=evening}      => {Item=Coffee} 0.004242454  0.1673077
0.02535720 0.6271210    87
```

```
inspect(head(sort(caf_rule[bread], decreasing = TRUE),by = "lift",10))
```

```
##      lhs                           rhs              support confidence
coverage      lift count
## [1]  {period_day=morning,
##       weekday_weekend=weekend} => {Item=Bread} 0.030379870  0.1928793
0.15750719 1.1895864   623
## [2]  {period_day=morning}      => {Item=Bread} 0.078509777  0.1915754
0.40981128 1.1815451  1610
## [3]  {period_day=morning,
##       weekday_weekend=weekday} => {Item=Bread} 0.048129907  0.1907615
0.25230409 1.1765251   987
## [4]  {weekday_weekend=weekday} => {Item=Bread} 0.102013946  0.1633482
0.62451846 1.0074529  2092
## [5]  {weekday_weekend=weekend} => {Item=Bread} 0.060125811  0.1601299
0.37548154 0.9876040  1233
## [6]  {period_day=afternoon,
##       weekday_weekend=weekday} => {Item=Bread} 0.051689667  0.1457445
0.35465938 0.8988822  1060
## [7]  {period_day=afternoon}    => {Item=Bread} 0.080996733  0.1435733
0.56414883 0.8854913  1661
## [8]  {period_day=afternoon,
##       weekday_weekend=weekend} => {Item=Bread} 0.029307066  0.1398976
0.20948944 0.8628209   601
## [9]  {period_day=evening,
##       weekday_weekend=weekday} => {Item=Bread} 0.002194373  0.1264045
0.01735993 0.7796021    45
## [10] {period_day=evening}      => {Item=Bread} 0.002633247  0.1038462
0.02535720 0.6404731    54
```

```
inspect(head(sort(caf_rule[tea], decreasing = TRUE),by = "lift",10))
```

```
##      lhs                           rhs            support confidence
coverage      lift count
## [1]  {period_day=evening,
##       weekday_weekend=weekday} => {Item=Tea} 0.001950553 0.11235955
0.01735993 1.6056845    40
## [2]  {period_day=evening}      => {Item=Tea} 0.002389428 0.09423077
0.02535720 1.3466135    49
## [3]  {period_day=afternoon,
##       weekday_weekend=weekday} => {Item=Tea} 0.029648413 0.08359687
0.35465938 1.1946487   608
```

```
## [4]  {period_day=afternoon}    => {Item=Tea} 0.045350368 0.08038724
0.56414883 1.1487813   930
## [5]  {weekday_weekend=weekday} => {Item=Tea} 0.047593505 0.07620832
0.62451846 1.0890621   976
## [6]  {period_day=afternoon,
##       weekday_weekend=weekend} => {Item=Tea} 0.015701955 0.07495345
0.20948944 1.0711291   322
## [7]  {period_day=morning,
##       weekday_weekend=weekday} => {Item=Tea} 0.015994538 0.06339389
0.25230409 0.9059363   328
## [8]  {weekday_weekend=weekend} => {Item=Tea} 0.022382601 0.05961039
0.37548154 0.8518678   459
## [9]  {period_day=morning}      => {Item=Tea} 0.022236310 0.05425988
0.40981128 0.7754058   456
## [10] {period_day=morning,
##       weekday_weekend=weekend} => {Item=Tea} 0.006241771 0.03962848
0.15750719 0.5663145   128
```

coffee has most relation in afternoon on a weekday with highest Lift and Confidence bread has most relation in morning on weekday with highest Lift and Confidence tea has most relation in afternoon on weekday with highest Lift and Confidence

```
top_3 <- c(coffee, bread, tea)

cafe <- read.csv('bread+basket.csv', sep=',')
cafe_tran_item <- as.matrix(table(cafe$Transaction, cafe$Item))
cafe_tran_item <- (cafe_tran_item==1)

cafe_rules_1 <- apriori(cafe_tran_item, parameter =
list(minlen=2,maxlen=2,support = 0.001, confidence = 0.001))

## Apriori
##
## Parameter specification:
##  confidence minval smax arem  aval originalSupport maxtime support minlen
##       0.001    0.1    1 none FALSE            TRUE       5   0.001      2
##  maxlen target  ext
##       2  rules TRUE
##
## Algorithmic control:
##  filter tree heap memopt load sort verbose
##     0.1 TRUE TRUE  FALSE TRUE    2    TRUE
##
## Absolute minimum support count: 9
##
## set item appearances ...[0 item(s)] done [0.00s].
## set transactions ...[94 item(s), 9465 transaction(s)] done [0.00s].
## sorting and recoding items ... [57 item(s)] done [0.00s].
## creating transaction tree ... done [0.00s].
## checking subsets of size 1 2
```

```
## Warning in apriori(cafe_tran_item, parameter = list(minlen = 2, maxlen =
2, :
## Mining stopped (maxlen reached). Only patterns up to a length of 2
returned!

##  done [0.00s].
## writing ... [476 rule(s)] done [0.00s].
## creating S4 object  ... done [0.00s].
```

```
coffee <- 1:length(cafe_rules_1)
coffee <-
coffee[str_detect(cafe_rules_1@rhs@itemInfo$labels[cafe_rules_1@rhs@data@i+1]
,"Coffee")]

bread <- 1:length(cafe_rules_1)
bread <-
bread[str_detect(cafe_rules_1@rhs@itemInfo$labels[cafe_rules_1@rhs@data@i+1]
,"Bread")]

tea <- 1:length(cafe_rules_1)
tea <-
tea[str_detect(cafe_rules_1@rhs@itemInfo$labels[cafe_rules_1@rhs@data@i+1]
,"Tea")]

top_3 <- c(coffee, bread, tea)


inspect(head(sort(cafe_rules_1[-top_3], decreasing = TRUE,by = "lift"),10))
```

```
##      lhs                         rhs                     support
confidence
## [1]  {Extra Salami or Feta} => {Salad}               0.001690438
0.42105263
## [2]  {Salad}                => {Extra Salami or Feta} 0.001690438
0.16161616
## [3]  {Fudge}                => {Jam}                 0.001796091
0.13385827
## [4]  {Jam}                  => {Fudge}               0.001796091
0.12592593
## [5]  {Spanish Brunch}       => {Salad}               0.001267829
0.06976744
## [6]  {Salad}                => {Spanish Brunch}      0.001267829
0.12121212
## [7]  {Jammie Dodgers}       => {Tiffin}              0.001162176
0.08800000
## [8]  {Tiffin}               => {Jammie Dodgers}      0.001162176
0.07534247
## [9]  {Chicken Stew}         => {Spanish Brunch}      0.001162176
0.08943089
## [10] {Spanish Brunch}       => {Chicken Stew}        0.001162176
```

```
0.06395349
##      coverage    lift      count
## [1]  0.004014791 40.255183 16
## [2]  0.010459588 40.255183 16
## [3]  0.013417855  9.384952 17
## [4]  0.014263074  9.384952 17
## [5]  0.018172213  6.670190 12
## [6]  0.010459588  6.670190 12
## [7]  0.013206550  5.704932 11
## [8]  0.015425251  5.704932 11
## [9]  0.012995246  4.921299 11
## [10] 0.018172213  4.921299 11
```

highest relation with Lift : {Extra Salami or Feta} => {Salad} and {Salad} => {Extra Salami or Feta} that's mean customer who order salad will prefer "Extra Salami or Feta" as the topping more than the other such as Soup or Spanish Brunch

```
salad <- 1:length(cafe_rules_1)
salad <-
salad[str_detect(cafe_rules_1@rhs@itemInfo$labels[cafe_rules_1@rhs@data@i+1]
,"Salad")]
inspect(head(sort(cafe_rules_1[salad], decreasing = TRUE,by = "lift"),10))

##     lhs                        rhs     support     confidence coverage
## [1] {Extra Salami or Feta} => {Salad} 0.001690438 0.42105263 0.004014791
## [2] {Spanish Brunch}       => {Salad} 0.001267829 0.06976744 0.018172213
## [3] {Soup}                 => {Salad} 0.001162176 0.03536977 0.032857898
## [4] {Juice}                => {Salad} 0.001056524 0.02770083 0.038140518
## [5] {Sandwich}             => {Salad} 0.001690438 0.02693603 0.062757528
## [6] {Tea}                  => {Salad} 0.001796091 0.01340694 0.133967248
## [7] {Coffee}               => {Salad} 0.004648706 0.01203501 0.386265188
## [8] {Cake}                 => {Salad} 0.001162176 0.01168969 0.099418912
## [9] {Bread}                => {Salad} 0.001901743 0.00625000 0.304278922
##     lift       count
## [1] 40.2551834 16
## [2]  6.6701903 12
## [3]  3.3815648 11
## [4]  2.6483673 10
## [5]  2.5752474 16
## [6]  1.2817847 17
## [7]  1.1506200 44
## [8]  1.1176054 11
## [9]  0.5975379 18

salami <- 1:length(cafe_rules_1)
salami <-
salami[str_detect(cafe_rules_1@rhs@itemInfo$labels[cafe_rules_1@rhs@data@i+1]
,"Salami")]
inspect(head(sort(cafe_rules_1[salami], decreasing = TRUE,by = "lift"),10))
```

```
##     lhs         rhs                     support     confidence coverage
## [1] {Salad}  => {Extra Salami or Feta} 0.001690438 0.16161616 0.01045959
## [2] {Coffee} => {Extra Salami or Feta} 0.002113048 0.00547046 0.38626519
##     lift      count
## [1] 40.255183 16
## [2]  1.362576 20
```

In term of Business : To increase sales is to attach the Menu with the Image of this three combination “Salad with Extra Salami or Feta beside with coffee” for remind customer.